MIL-P-60029B (MU)
20 June 1973
SUPERSEDING
MIL-P-60029A (MU)
29 April 1965

MILITARY SPECIFICATION

PROPELLANT, M9 FOR USE IN INCREMENTS M90 AND M90A1 FOR 81MM MORTAR

1. SCOPE

1.1 Scope.-This specification covers two types of flake propellant designated as M9 for use in charge A and B for the M90 and M90A1 increments for the following 81MM cartridges: M374, M374A2, M375 and M375A2.

1.2 Classification.-The propellant shall be of the following types: Increment Charge A to be used at Zone 1 and Increment Charge B at Zones 2 through 9.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-652 - Propellants Solid for Cannons Requirements and Packing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1171 - Acceptance and Description Sheets for Propellants and Explosives

FSC: 1376

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DRAWINGS

ARMY

8881021 - Charge A, Propellant Increment M90, Mortar Cartridge, 81MM
8881023 - Charge B, Propellant Increment M90, Mortar Cartridge, 81MM
9233369 - Charge A, Propellant Increment M90A1, Mortar Cartridge, 81MM
9233371 - Charge B, Propellant Increment M90A1, Mortar Cartridge, 81MM

Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Transportation: Parts 171-179

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington,D.C. 20402. Orders for the above publication should cite: 49 CFR 171-179 (latest revision).

3. REQUIREMENTS

3.1 Material.-The propellant shall be M9 flake propellant conforming to the requirements of MIL-STD-652 and either Drawings 8881021 and 8881023 or 9233369 and 9233371.

3.2 Proving Ground.-The propellant, when fired in a standard weapon and with the weight of projectile (9.12 pounds (lbs)) for which it was manufactured, shall comply with the following requirements when tested as specified in 4.3.

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3.2.1 Charge Assessment.-At 70 degrees Fahrenheit (°F) and standard weapon conditions, the propellant shall be capable of charge assessment to a velocity of 344 feet per second (fps) at Zone 1 and 866 feet per second (fps) at Zone 9 without an associated expected pressure greater than 9200 pounds per square inch (psi).

3.2.2 Velocity uniformity.-The lot standard deviation of velocity shall not exceed the requirements of Table I when determined as specified in 4.2.2.2.

TABLE I

Zone No.	Maximum Standard Deviation fps
1	2.1
9	3.4

3.2.3 Permissible Individual Maximum Pressure.-The propellant at recommended charge weight shall not produce a chamber pressure greater than 10,600 psi at any operating temperature (-50°F to +145°F) when fired in a standard weapon.

3.3 Charge Weight

3.3.1 Charge A.-The actual loading weight shall be determined by assessment. The assessed weight shall not be greater than 187 grains.

3.3.2 Charge B.-The actual loading weight shall be determined by assessment. The assessed weight shall not be greater than 171 grains.

3.4 Graphite Glaze.-The graphite glaze shall not exceed 0.40 percent on an added basis when tested as specified in accordance with Method 308.1 of MIL-STD-286.

3.5 Workmanship.-The best commercial practices shall be used in the formulation of propellant furnished under this specification and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

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4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109, in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of batches of product inspected (shall be made available upon request by the contracting officer)

c. Results obtained, by defect code, for all inspections performed (shall be made available upon request by the contracting officer)

d. Drawing, specification number, and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contracting documents (shall be made available upon request by the contracting officer).

f. Number of items in the lot,

g. Date submitted.

h. Eight copies of propellant description sheet SMU Form 1060 containing the applicable data (see 6.8).

4.1.2 The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.2 Inspection provisions

4.2.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot which is defined as an essentially homogeneous quantity of propellant from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. A lot shall consist of one or more sub-lots of M9 propellant produced by one manufacturer, in accordance with the same specification, and same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of M9 propellant which has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

4.2.2 Testing

4.2.2.1 Materials (see 3.1).-Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical-physical properties in accordance with MIL-STD-652.

4.2.2.2 Proving Ground (see 3.2).-Each lot of propellant shall be sampled for ballistic acceptance in accordance with MIL-STD-652

4.2.2.2.1 Charge Assessment (see 3.2.1).-Major defect, Code No. 01001 (see 6.4).-The lot shall be rejected if the assessed charge weight of propellant has an expected pressure greater than 9,200 psi when tested in accordance with 4.3.2.

4.2.2.2.2 Velocity Uniformity.-(see 3.2.2) Major defect, Code No. 02001.-If the product of the factor times the velocity standard deviation of rounds tested in accordance with 4.3.3 exceeds the applicable requirement, the lot shall be rejected (see 6.5)

4.2.2.2.3 Individual pressure (see 3.2.3)

4.2.2.2.3.1 Initial production

4.2.2.2.3.1.1 145°F.-If any individual pressure observed during uniformity firings at 145°F after correction for difference between as fired and recommended service charge weight exceeds 10,600 psi, the lot shall be rejected.

4.2.2.2.3.1.2 70°F.-If any individual pressures observed during the 70°F uniformity series when corrected to standard conditions and recommended charge weight exceeds 10,600 psi the lot shall be rejected.

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4.2.2.2.3.1.3 -50°F.-If any individual pressure observed during uniformity firings at -50°F after correction for difference between as fired and recommended service charge weight exceeds 10,600 psi, the lot shall be rejected.

4.2.2.2.3.2 Subsequent Production 70°F.-If any individual pressures observed during the 70°F uniformity series when corrected to standard conditions and recommended charge weight exceeds 10,600 psi minus 75 times the pressure temperature coefficient, (psi/°F), established during Initial Production Testing, the lot shall be tested at +145°F in accordance with 4.3.3.1.1.

4.2.2.2.4 Charge Weight (see 3.3)Major defect, Code No. 03001.-The propellant lot shall be rejected if the assessed charge weight exceeds the applicable requirements.

4.2.2.3 Graphite glaze (see 3.4) Major defect, Code No. 04001.-The propellant lot shall be rejected if the graphite glaze exceeds 0.40 percent on an added basis. When tested in accordance with Method 308.1 of MIL-STD-236. If the difference between the final recommended charge weight and the charge weight employed in the uniformity series exceeds 5 grains, the ten round uniformity test shall be rejected with the recommended charge weight.

4.3 Test methods and procedures

4.3.1 Proving ground tests.-These tests shall be conducted at a Government proving ground (designated by the contracting officer) in accordance with the applicable Proving Ground Acceptance Test Procedure. Tests for charge assessment and uniformity shall be performed using increment bags fabricated from the same material with which the propellant lot is intended for use.

4.3.2 Charge assessment

4.3.2.1 Charge Slope Method.-Three or more groups of rounds loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 70°F. From the data obtained, a charge weight shall be calculated which will yield the prescribed service velocity. This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.3.3). The difference between the

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corrected velocity obtained during the uniformity series and the service velocity shall be compensated for by a final adjustment in charge weight using the slope of the velocity charge weight curve previously established.

4.3.2.2 Constant Slope Method.-After ten (10) lots of a given production series have been fired to provide a good knowledge of the propellant characteristics, the charge slope method of assessment may be discontinued and the charge weight determined by the constant slope method. A composite velocity versus charge weight versus pressure curve can be established from test results generated. This curve can then be used as in 4.3.2.1 above to calculate the charge weight needed to achieve service velocity. This calculated charge weight is then to be loaded into rounds for the uniformity series and final adjustment is to be made as indicated above for the charge slope method.

4.3.3 Uniformity series

4.3.3.1 Initial production

4.3.3.1.1 145°F.-For each increment level, A or B ten rounds of M374A2 Cartridges shall be assembled with the assessed charge of propellant, conditioned to 145°F (as indicated by a thermocoupled control round (rnd) and fired alternately with calibration rounds. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to recommended charge weight conditions (i.e. difference between as fired charge weight and recommended charge weight). (see 6.10)

4.3.3.1.2 70°F.-For each increment level, A or B ten rounds of M374A2 Cartridges shall be assembled with the assessed charge of propellant, conditioned to 70°F (as indicated by thermocoupled control rnd) and fired alternately with calibration rounds. The velocity and pressure means, as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to standard shell weight (9.12 lbs) and standard weapon/firing conditions (i.e. the difference between calibration performance under test conditions and standard conditons) and to recommended charge weight conditions.

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4.3.3.1.3 -50°F.-For each increment level, A or B ten rounds of M374A2 Cartridges shall be assembed with the assessed charge of propellant, conditioned to -50°F (as indicated by a thermocoupled control rnd) and fired alternately with calibration rounds. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results shall be corrected to recommended charge weight conditions (i.e. difference between as fired charge weight and recommdned charge weight).

4.3.3.2 Subsequent Production.-For each increment level, A or B seven rounds of M374A2 Cartridges shall be assembled and tested at 70°F while 145°F and -50°F testing shall be omitted entirely. The velocity and pressure means as well as the standard deviations associated with these parameters shall be calculated. All results at 70°F shall be corrected to standard shell weight (9.12 lbs) and standard weapon/firing conditions and recommended charge weight conditions (see 4.2.2.2.3.2).

4.3.4 Refiring.-If for any reason the proving ground or technical agency considers that test conditions have detrimentally affected the test result, additional charges as required shall be tested.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with MIL-STD-652.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1)
c. Process. Details of the manufacturing process and the euqipment used by the Contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.

6.2 Standard deviation.-The standard deviation shall be calculated with (n-1) as the devisor in a standard statistical technique equivalent to that shown in MIL-STD-414, Section B, Example B-1.

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6.3 Submission of test data.-In addition to the normal distribution of records, when the propellant is procured by the Department of the Army, one copy of all ballistic test data shall be forwarded to: Commander, Picatinny Arsenal, ATTN: SMUPA-QA-A-P, and one copy to ATTN: SMUPA-ADEP-2, Dover, New Jersey 07801.

6.4 Inspection code number.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.5 Factors.-The factors, as taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limit (A.05) and n-1 degrees of freedom, make allowances for the probability that a standard deviation from a smaller sample size exceeds the true standard deviation by chance alone.

6.6 Initial production lots.-The initial production lots represent the first three consecutive lots from each manufacturer which have been produced and met the acceptance criteria of this specification. Lapse in production of 90 days or more shall cause testing to revert back to initial production plan.

6.7 Tolerance.-A tolerance of -5°F applies to all -50°F and 145°F requirements and a tolerance of plus or minus 2°F to all 70°F requirements.

6.8 Submission of SMU Form 1060.-Information contained in propellant description sheet, SMU Form 1060, need not be resubmitted in the form of a certificate provided the provisions regarding certifying of responsible agents is adhered to (see 4.1.2).

6.9 Surveillance sample.-Samples shall be forwarded to Picatinny Arsenal, Dover, N.J. ATTN: SMUPA-FR-G, for the propellant surveillance test program in accordance with MIL-STD-652.

6.10 Test cartridges shall be assembled in accordance with the following drawings:

9240950 - Cartridge, 81MM: HE, M374A2 w/Fuze PD 524A6

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6.11 Proving ground test summary

Initial Production

Temp (°F)	Sample Size	Vel (fps) (assessed	Lot Std. Dev of Vel (fps)	Perm. Ind. Max Pres (psi)	Avg. Pres. Max (psi)
-50	10	NA	NA	10,600	NA
70	10	344-Zone 1 866-Zone 9	2.6-Zone 1 3.4-Zone 9	10,600	9200
145	10	NA	NA	10,600	NA

Subsequent Production

70	7	344-Zone 1 866-Zone 9	2.6-Zone 1 3.4-Zone 9	10,600	9200
145*	10	NA	NA	10,600	NA

*To be fired only when 70°F results indicate that marginal pressure performance may exist (see 4.2.2.2.3.2).

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1376-A-011

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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